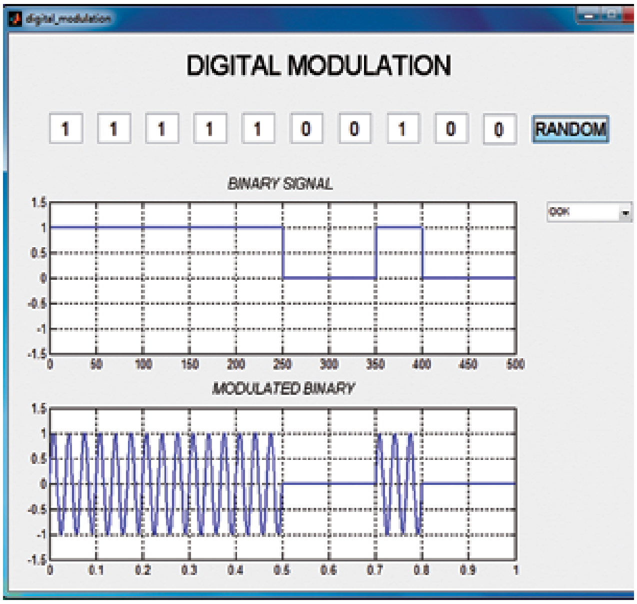

Fig. 5: OOK modulation

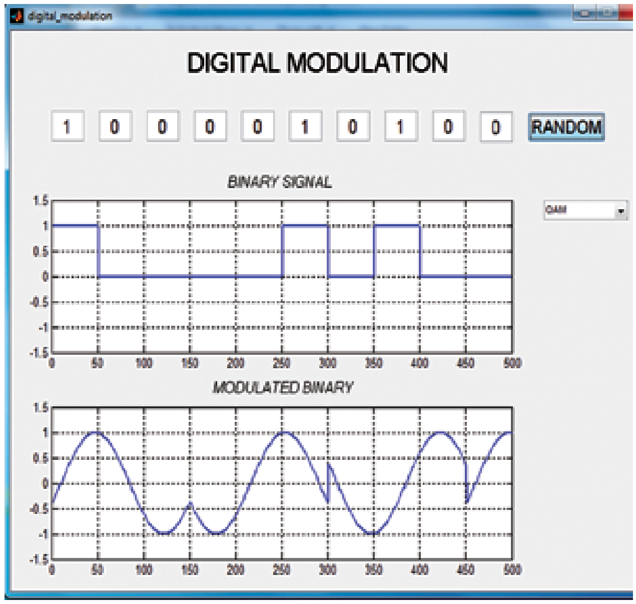

Fig. 6: QAM modulation

BPSK modulation, digital data of 1 and 0 are represented by 180-degree phase change. Here, one bit per symbol is used. In QPSK modulation these are represented by phase shift of 90 degrees and so two bits are mapped on each signal. In multilevel PSK more than two bits are mapped using different phase angles. In 8PSK eight different phase angles are used where the incoming bits are encoded using three bits. Fig. 3 shows the simulation output of 8PSK signal.

ASK modulation

ASK is a type of amplitude modulation which represents binary data in the form of variations in the amplitude of a signal. Any modulated signal has high-frequency carrier signal. In ASK modulation, the level of signal amplitude can be used to represent binary logic 0's and 1's. Thus, logic 0 is represented by the lower level amplitude of carrier signal and logic 1 by the higher amplitude of carrier signal as shown in Fig. 4.

OOK modulation

OOK modulation is a popular technique used in control applications because of its simplicity and low implementation cost. It has the advantage of allowing the transmitter to idle during the transmission of a '0,' thus conserving power. The disadvantage of OOK modulation lies in the presence of an undesired signal. As the proliferation of control and data communication apparatus increases, so does the aggravation of not being able to communicate.

OOK is a form of ASK modulation that represents digital data as the presence or absence of a carrier signal. We can think of carrier signal as an on or off switch. In its simplest form, the presence of a carrier signal represents a binary '1' and its absence represents a binary '0' as shown in Fig. 5.

QAM modulation

QAM has been used for some analogue transmissions including AM stereo transmissions. QAM provides a highly effective form of modulation for data and is used in cellular phones and Wi-Fi systems.

QAM is a signal in which two carriers shifted in phase by 90 degrees (that is, sine and cosine) are modulated and combined. As a result of their 90° phase difference, they are in quadrature and hence the name. Often one signal is called the in-phase or 'I' signal and the other the quadrature or 'Q' signal. The modulated output waveform obtained from MATLAB simulation is shown in Fig. 6.

The resultant overall signal consisting of the combination of both I and Q carriers contains both amplitude and phase variations. In view of the fact that both amplitude and phase variations are present, it may also be considered as a mixture of amplitude and phase modulations. QAM is the name of a family of digital modulation methods and a related family of analogue modulation methods widely used in modern telecommunications to transmit information.

It conveys two analogue message signals, or two digital bit streams by changing (modulating) the amplitudes of two carrier waves. QAM is used extensively as a modulation scheme for digital telecommunication systems such as in 802.11 Wi-Fi standards. Arbitrarily high spectral efficiencies can be achieved with QAM by setting a suitable constellation size, limited only by the noise level and linearity of the communications channel. QAM is being used in optical fibre systems. EFY

EFY Note
The source code of this project is available for free download at *source.efymag.com*

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